



**The University of Azad Jammu & Kashmir,
Muzaffarabad**

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Roll NO: 2024-SE-06

Semester: 4th

Course Title: Computer Networks

Course Code: CS-2203

Lab NO: 11

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✓ Open Ended Lab

1. Network Requirements Analysis

The proposed educational institute contains three departments: Administration, Academic, and IT. The network is designed to provide reliable communication between departments while leaving room for future expansion.

Department	PCs	Switch	Network
Administration	5	SW1 – 24-port	192.168.10.0/24
Academic	8	SW2 – 24-port	192.168.20.0/24
IT	5	SW3 – 24-port	192.168.30.0/24
Total	18	3 switches	3 LANs

Additional equipment: one Cisco 2911 router, three Cisco 2960 24-port switches, 18 PCs, and Ethernet cables.

2. Topology Selection and Justification

Star topology is selected for each departmental LAN. Each PC connects to a central switch, and the three switches connect to a central router.

- Easy installation and administration.
- Failure of one PC or cable does not normally affect other PCs.
- Easy troubleshooting and expansion.
- Good performance for a small institute network.
- Separate departmental LANs can communicate through the router.

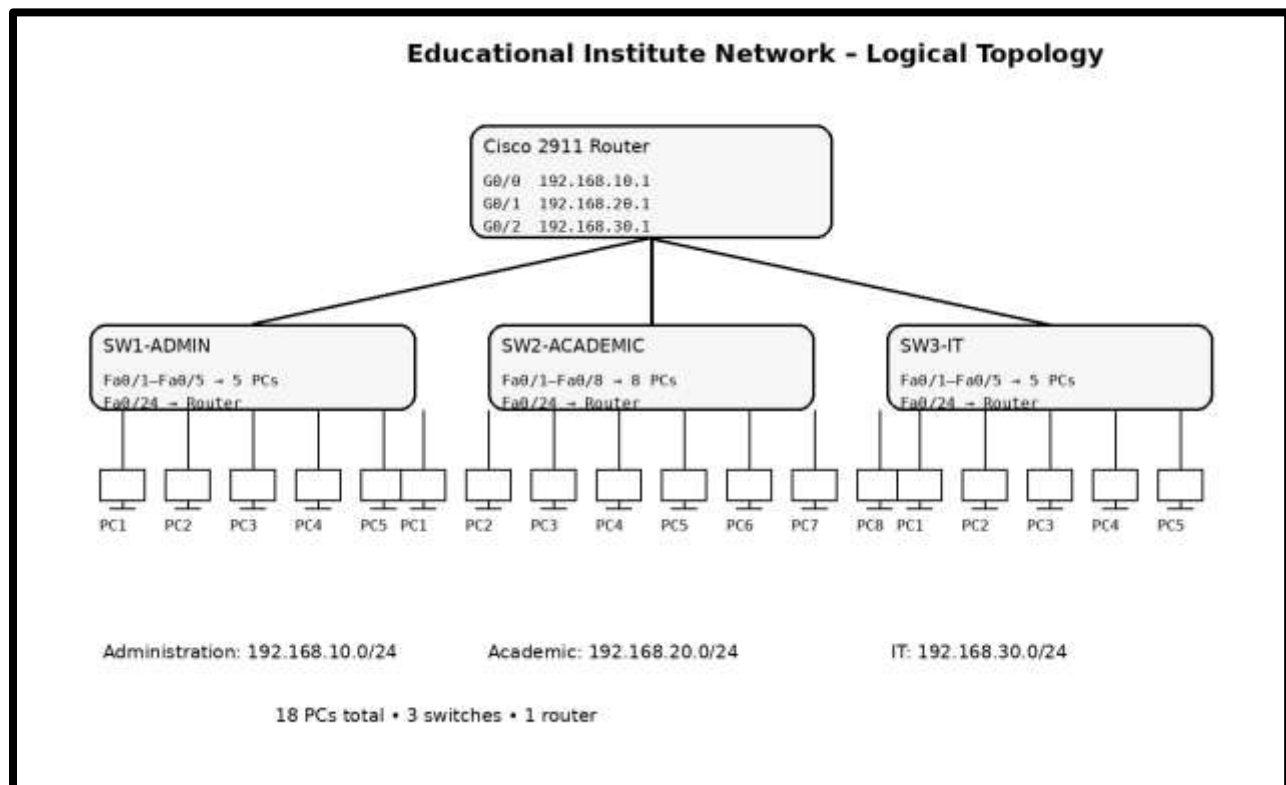
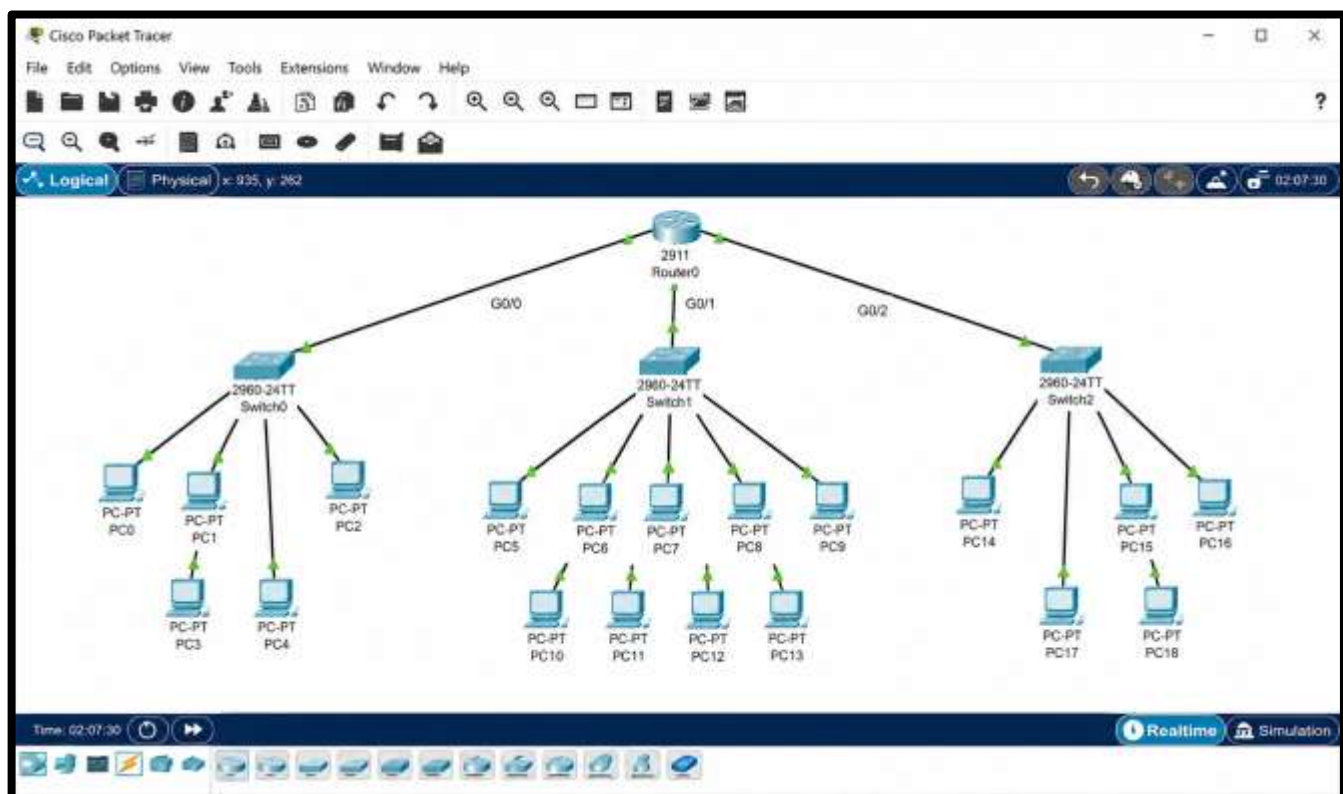


Figure 1: Logical network diagram

➤ Implementation On Cisco Packet Tracer



3. Network Cabling Plan

Connection	Recommended cable
PC → Switch	Copper Straight-Through Ethernet
Switch → Router	Copper Straight-Through Ethernet
Future switch-to-switch connection	Copper Crossover or Auto-Connect, depending on device/interface

Copper Ethernet is appropriate for the small LAN because it is economical, easy to install, and supported by the Packet Tracer devices used in this lab.

4. Switch Design

Three Cisco 2960 switches are used. Each switch has enough unused ports for future devices.

Switch	Department	PC ports	Router uplink
SW1-ADMIN	Administration	Fa0/1–Fa0/5	Fa0/24
SW2-ACADEMIC	Academic	Fa0/1–Fa0/8	Fa0/24
SW3-IT	IT	Fa0/1–Fa0/5	Fa0/24

Basic Switch Configuration

The following commands can be entered on each switch:

```
enable
configure terminal
hostname SW1-ADMIN
enable secret class123
end
write memory
```

For SW2 and SW3, change the hostname to SW2-ACADEMIC and SW3-IT respectively.

5. IP Addressing Scheme

A separate /24 private IPv4 network is assigned to each department. The router interface in each subnet acts as the default gateway.

Department	Network	Gateway	Example PC range
Administration	192.168.10.0/24	192.168.10.1	192.168.10.10–14
Academic	192.168.20.0/24	192.168.20.1	192.168.20.10–17
IT	192.168.30.0/24	192.168.30.1	192.168.30.10–14

6. Detailed IP Addressing Table

Device	IP Address	Subnet Mask	Default Gateway
Admin PC1	192.168.10.10	255.255.255.0	192.168.10.1
Admin PC2	192.168.10.11	255.255.255.0	192.168.10.1
Admin PC3	192.168.10.12	255.255.255.0	192.168.10.1
Admin PC4	192.168.10.13	255.255.255.0	192.168.10.1
Admin PC5	192.168.10.14	255.255.255.0	192.168.10.1
Academic PC1	192.168.20.10	255.255.255.0	192.168.20.1
Academic PC2	192.168.20.11	255.255.255.0	192.168.20.1
Academic PC3	192.168.20.12	255.255.255.0	192.168.20.1
Academic PC4	192.168.20.13	255.255.255.0	192.168.20.1
Academic PC5	192.168.20.14	255.255.255.0	192.168.20.1
Academic PC6	192.168.20.15	255.255.255.0	192.168.20.1
Academic PC7	192.168.20.16	255.255.255.0	192.168.20.1
Academic PC8	192.168.20.17	255.255.255.0	192.168.20.1
IT PC1	192.168.30.10	255.255.255.0	192.168.30.1
IT PC2	192.168.30.11	255.255.255.0	192.168.30.1
IT PC3	192.168.30.12	255.255.255.0	192.168.30.1
IT PC4	192.168.30.13	255.255.255.0	192.168.30.1
IT PC5	192.168.30.14	255.255.255.0	192.168.30.1

7. Router Configuration

The router uses three Gigabit Ethernet interfaces. Each interface is assigned the gateway address for one departmental subnet.

Router CLI - Configuration Commands

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 192.168.20.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/2
Router(config-if)# ip address 192.168.30.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# end
Router# write memory
```

Figure 2: Router configuration command

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
interface gigabitEthernet 0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
interface gigabitEthernet 0/2
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
end
write memory
```

8. PC Configuration

On each PC, open Desktop → IP Configuration → Static. Enter the IP address, subnet mask 255.255.255.0, and the appropriate default gateway from the IP addressing table.

Example: Administration PC1 = 192.168.10.10, mask = 255.255.255.0, gateway = 192.168.10.1.

9. Connectivity Testing

After configuring the devices, use the PC Command Prompt to test communication. For example, from Administration PC1:

```
ping 192.168.20.10
```

A successful result should show replies with 0% packet loss. Also test 192.168.30.10 and the local gateway 192.168.10.1.

Connectivity Test – Expected Packet Tracer CLI Output

```
PC> ping 192.168.20.10
Pinging 192.168.20.10 with 32 bytes of data:
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Ping statistics: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

10. Future Improvements

- Implement VLANs to logically separate Administration, Academic, and IT traffic.
- Deploy a DHCP server to automatically assign IP addresses.
- Add wireless access points for laptops and mobile devices.
- Install a firewall between the internal network and the Internet.
- Add redundancy using additional links/devices as the institute grows.
- Implement network monitoring and centralized logging.

11. Conclusion

The proposed network provides a simple, scalable, and manageable design for a small educational institute. Star topology, three departmental LANs, a central router, 24-port switches, and IPv4 /24 subnets provide the required connectivity while leaving significant room for expansion. Connectivity can be verified using ping tests and router interface status commands.
